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Applied Genomics

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Cell line characterization - Clonality analysis AP-CLCH-CLA

Applications

Cell line authentication is essential to ensure research validity, reproducibility, and data integrity. Genetic characterization of cell lines using sequencing, enabling reliable identification, authentication, and detection of cross-contamination. The method involves use of proprietary process to analyze the insertional; sites of transgene construct in complex genomes such as CHO. The process enables identification of insertion copies. Identification of transgene insert integrity and ensure Master Cell Bank/ Working Cell Bank clones encode mutations free proteins. The process also uses whole genome SNP and CNV analysis to match the host genome and ensure it is free of any viral integration events.
The report can provide data for filing IP on clones, provide characterization for regulatory filing.

Pros

- High accuracy and reproducibility
- Rapid and cost-effective
- Detects cross-contamination
- Suitable for routine QC
- Low DNA input required

Notes

- Limited to targeted loci
- Reference profiles required
- Not genome-wide
- Periodic testing recommended

Input

Genomic DNA

- DNA Quantity: ≥ 100 ng - 1µg (recommended)
- High Molecular Weight DNA
- Requires information of vectors used for transgenesis

Sample Types

Human Cell-lines | Animal Cell-lines | Fresh or frozen cell pellets

Platform

ONT Long-read Sequencing Platforms

Process Map

Sample DNA Extraction > DNA QC > Library Prep > NGS Data Acquisition > Data QC > Cellline Characterization Modules > Reporting

Standard Deliverables

Download Sample Report

- Reporting Summary of method / Reference Genomes
- Site of integration of Transgenes
- Host genome CNV / SNP based similarity map
- Copy number of transgene- Count of sites of integration
- Abbrent Integration- Mutation/ Indels/ Truncation/ Deletion

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Hybridoma sequencing

AP-HYBM-SEQ

Applications

The hybridoma sequencing application aims to recover heavy and light chains from mouse, rat, and rabbit hybridoma. Hybridomas can encode multiple antibody transcripts, and in many instances express aberrant chain or multiple forms.
We offer chain annotation, Flagging of aberrant chains and provide add-on services to predict fuctional pairs using in-house data models, including gene-synthesis to re-construct the antibody V_H and V2 chains for expression.
This service focuses on sequencing antibody variable domains (VH and VL) hybridoma samples. By combining targeted amplification with next-generation sequencing, it enables reliable recovery of functional antibody sequences for cloning, expression, humanization, and intellectual property protection.
Hybridoma sequencing ensures long-term preservation of antibody sequence information independent of cell line stability.

Pros

- Accurate VH/VL sequence recovery
- Supports antibody cloning / Expression workflow
- Preserves antibody IP / Engineering

Notes

- Requires high-quality RNA/DNA
- Mixed clones may complicate analysis
- Recovered clone will need functional authentication

Input

Total RNA

- RNA: 200 ng - 500 ng

Sample Types

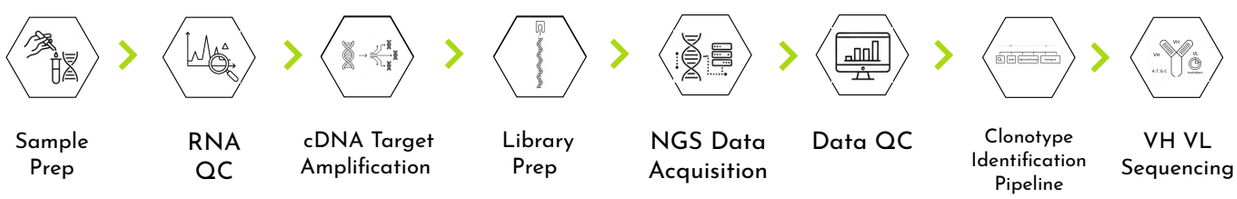
Hybridoma cell pellets

Cultured hybridoma cells

Platform

ONT Long-read sequencing platforms

Process Map



Standard Deliverables

Download Sample Report

- Sample QC Report, Summary of methods
- NGS Library summary
- VH / VL- Summary and sequence of chains
- Annotation of clonotypes/ flagging aberrant / non-productive chains

Add-ons

- In-Silico VH-VL pairing
- Gene reconstruction and synthesis
- Flagging aberrant chain
- In-Silico pairing
- Gene synthesis services

Contact us

Dr. Gopalakrishna Ramaswmy
Founder & CEO
Email: gk@theracues.com
Ph: +91 98805 22715

Dr. Sriram Parameswaran
Chief Scientific Officer
Email: sriram.parameswaran@theracues.com
Ph: +91 95910 06549

Theracues Innovations Pvt. Ltd.,
B Block, No. 1436, 2nd Floor,
Sahakar Nagar,
Bangalore, Karnataka 560092
Email: cues@theracues.com
Phone: +91 80 48661874

www.theracues.com

cues@theracues.com